

Long mutation cycles

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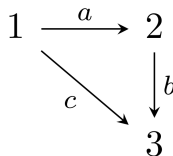
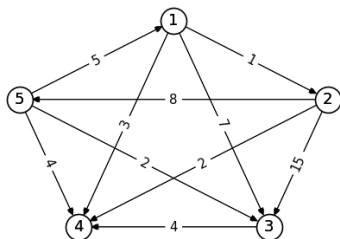
Seminar in Cluster algebras
MSU, April 2025

joint work with Sergey Fomin and Tucker Ervin
arxiv [2304.11505](#) and [2504.06573](#)

Quivers

Definition

A **quiver** is a finite oriented graph. Multiple edges are allowed, but no loops nor oriented 2-cycles. The multiplicities of arrows are called **weights**.



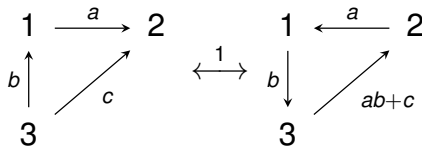
No **frozen** vertices!

Quiver mutation

Definition

Quiver **mutation** $\mu_i(Q) = Q'$ (or $Q \xleftrightarrow{i} Q'$) is computed in three steps.

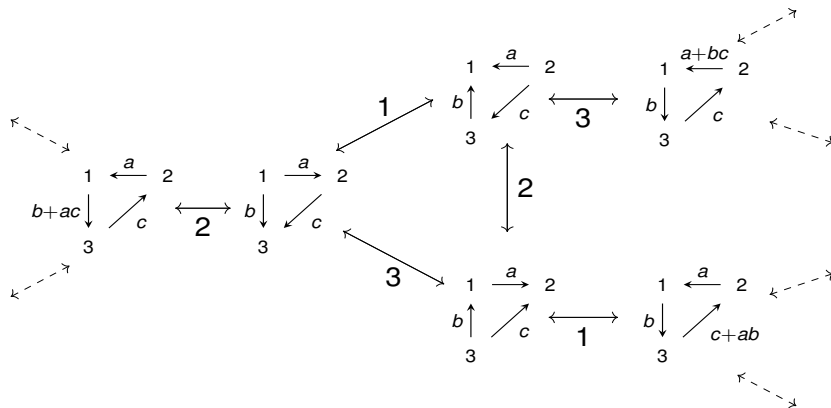
- If $u \xrightarrow{a} i \xrightarrow{b} v$ then add arrows $u \xrightarrow{ab} v$.
- Reverse all arrows incident to i .
- Delete oriented 2-cycles.



Mutation graph

Definition

The **mutation graph** of a mutation equivalence class $\mathbf{Q}$ is the graph with a vertex for each $Q \in \mathbf{Q}$ and an edge labeled i between Q and Q' whenever $\mu_i(Q) = Q'$.

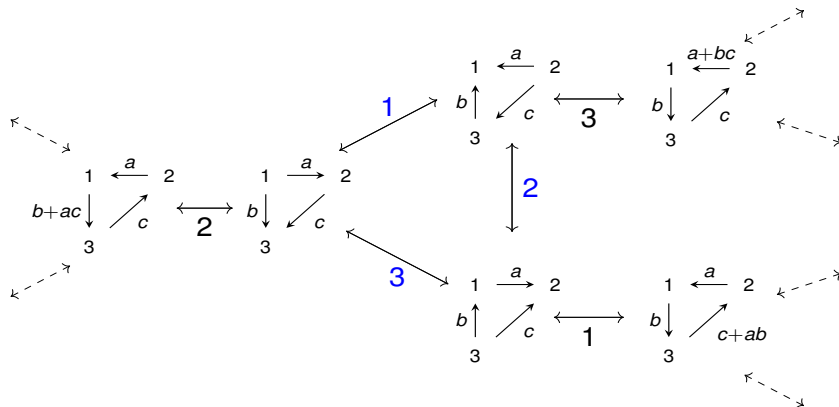


Mutation cycles

Definition

A **mutation cycle** of length $N > 0$ is defined by a quiver Q and a sequence $i_1 \cdots i_N$ such that $i_1 \neq i_2 \neq \cdots \neq i_{N-1} \neq i_N \neq i_1$ and

$$Q = Q_0 \xleftrightarrow{i_1} Q_1 \xleftrightarrow{i_2} \cdots \xleftrightarrow{i_{N-1}} Q_{N-1} \xleftrightarrow{i_N} Q_N = Q.$$



Mutation cycles

Example (Commuting mutations)

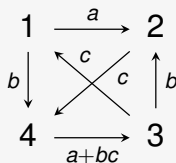
Commuting mutations produce mutation cycles of length 4.

Mutation cycles

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Example ([A.Fordy-B.R.Marsh, 2010])

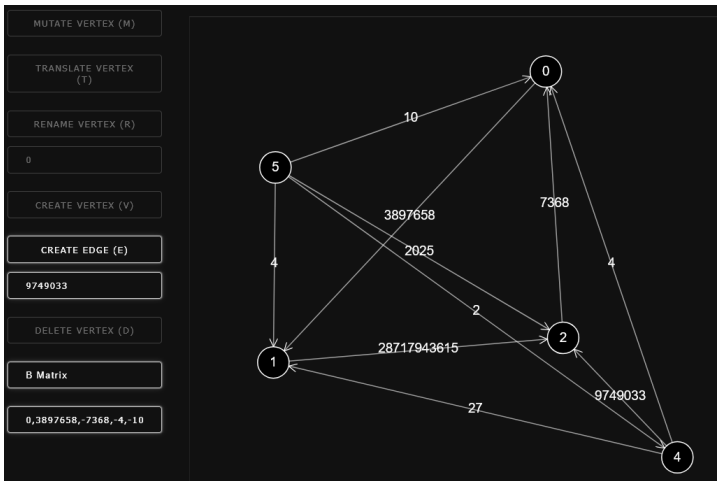


Applying $\mu_2 \circ \mu_3 \circ \mu_4 \circ \mu_1$ gives an isomorphic quiver ($\mu_4 \circ \mu_1$ rotates the quiver by 180 degrees)

Demonstration

Mutation cycles are not typical.

Webapp demonstration: Phylliida.dev/quiver.html



Main Theorem [Fomin-SN, 2023]

For any $n \geq 4$ and $k \geq 1$, we construct a family of n -vertex quivers Q such that:

- the weights of Q depend on an arbitrary choice of $\binom{n}{2}$ positive integer parameters;

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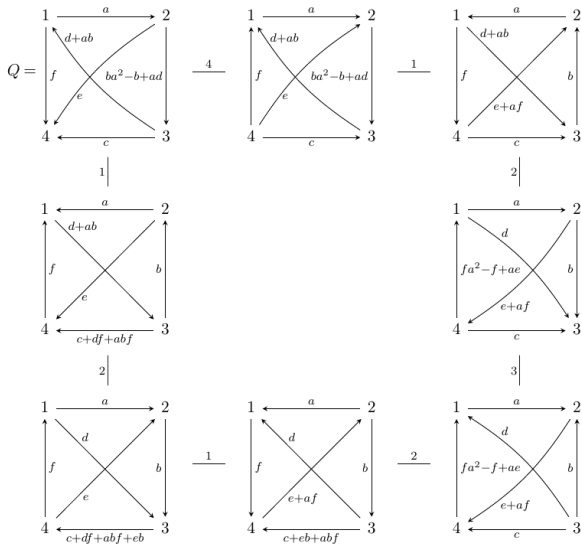
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Main Theorem [Fomin-SN, 2023]

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- the weights of Q depend on an arbitrary choice of $\binom{n}{2}$ positive integer parameters;
- Q lies on a mutation cycle of length $4k + n$;
- this cycle cannot be paved by mutation cycles of length $\leq 4k$.

Main theorem for $n = 4$ and $k = 1$



$$a, b, c, d, e, f \geq 2$$

Remark 1

For any given $n \geq 4$, our construction produces mutation cycles of arbitrarily large length.

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Remark 2

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Remark 3

These mutation cycles do not yield cycles in the exchange graph. Rather, they produce birational transformations that give rise to complicated discrete dynamical systems.

No pavings

Previously known long mutation cycles are all "paveable" by short cycles.

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Proposition

For $n=4$ and any k our mutation cycle is unique in its mutation class.

Not mutation-acyclic

Definition

A quiver is **mutation-acyclic** if it's mutation equiv. to an acyclic quiver.

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Theorem [Seven 2015]

The quivers we construct are not mutation-acyclic.

Acknowledgements:

- the construction (with Sergey Fomin)
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Definition

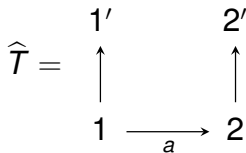
Let H, T be quivers. A **triangular extension** of H and T is the disjoint union of H and T , along with (any number of) new arrows directed from T to H .

$$T \xrightarrow{A} H$$

Reddening

The principle framing of T :

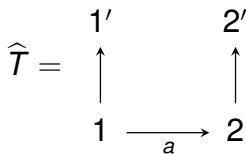
For each vertex i in T , add a new vertex i' and an arrow $i \rightarrow i'$ to form $\hat{T}$.



Reddening

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Definition [Keller 2013]

A mutation sequence $\mathbf{M}$ of T is a **reddening sequence** if every i' vertex is a source in $\mu_{\mathbf{M}}(\hat{T})$.

Making a mutation cycle

- 1 Choose two quivers H, T with reddening sequences $\mathbf{M}_H, \mathbf{M}_T$.

Making a mutation cycle

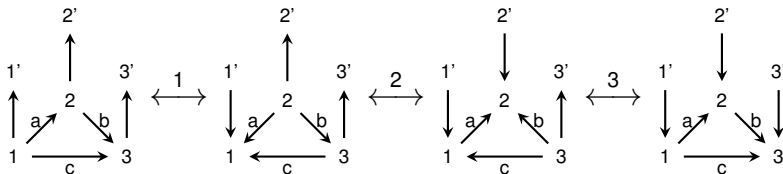
- 1 Choose two quivers H, T with reddening sequences $\mathbf{M}_H, \mathbf{M}_T$.
- 2 Create a triangular extension: $Q := T \xrightarrow{A} H$.

Theorem [Ervin-SN 2025]

The quiver Q is on a mutation cycle of length at least $|\mathbf{M}_T \mathbf{M}_H|$.

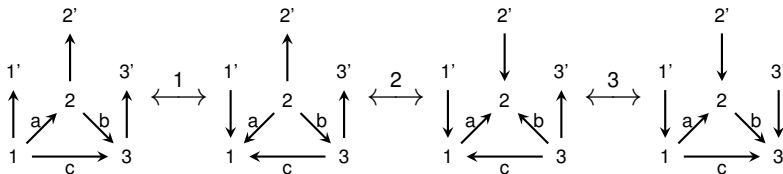
Reddening sequence examples

Example: source mutation cycles.

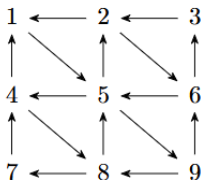


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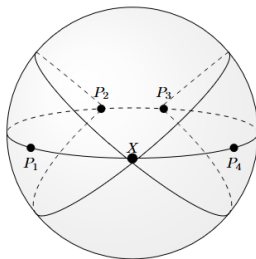


Example: Grassmannians. [Marsh-Scott 2016]



Reddening sequence 7, 4, 1, 8, 7, 5, 4, 2, 1, 9, 8, 7, 6, 5, 4, 3, 2, 1.

Reddening sequence examples



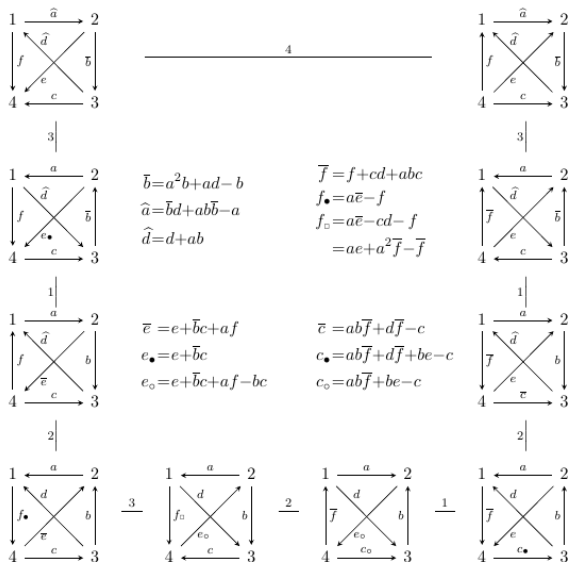
Theorem [Mills 2017]

If Σ is a marked surface which is not once-punctured and closed, then the quiver associated to any triangulation of Σ has a maximal green sequence.

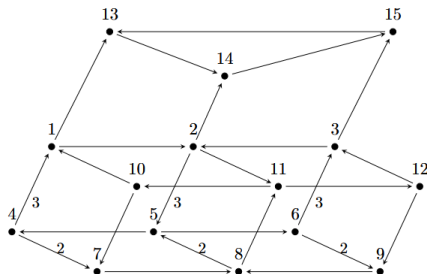
Theorem [Muller 2016]

If $\mathbf{M}$ is a mutation sequence and $\mathbf{S}$ is a reddening sequence of T then $\mathbf{M}^{-1}\mathbf{S}_{\sigma(\mathbf{M})}$ is a reddening sequence of $\mu_{\mathbf{M}}(T)$.

A 10-cycle



A quiver on many cycles



The quiver on $\{13, 14, 15\}$ lies on many reddening sequences of varying lengths, eg: 13, 14, 15, 13 and 13, 15, 13, 14, 13.

The complementary quiver has a reddening seq. of length 24 [Bucher et al. 2019].

Any triangular extension will lie on cycles of length 56 and 174 (and more!).

Two Lemmas

Lemma [Brüstle-Dupont-Pérotin 2014]

If L admits a reddening sequence $\mathbf{N}$ then $\mu_{\mathbf{N}}(L)$ is isomorphic to L .

Thus eventually $\mu_{\sigma^k(\mathbf{N})} \circ \cdots \circ \mu_{\sigma^0(\mathbf{N})}(L) = L$. If $\mathbf{N}$ is reduced, this is a mutation cycle.

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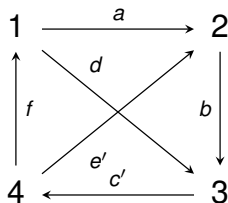
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Lemma [Cao-Li 2017 (with thanks to Fan Qin)]

If H and T are any two quivers with reddening sequences $\mathbf{M}, \mathbf{N}$ respectively, then the mutation sequence $\mathbf{NM}$ is a reddening sequence of any triangular extension $T \xrightarrow{A} H$.

Reddening sequences in triangular extensions are like source mutations in acyclic quivers.

A different mutation cycle of length 12



Pick integers $a, b, c, d, e, f \geq 2$ (with $d > cf, e > af$). Set

$$e' = e + bc - af,$$

$$c' = be + b^2c - c.$$

The quiver above is on a mutation cycle with mutation sequence

123234341412

More mutation cycles

We have constructed many other mutation cycles of 4-vertex quivers.

